

INT257:MODERN WEB APPLICATION DEVELOPMENT

L:2 T:0 P:2 Credits:3

Course Outcomes: Through this course students should be able to

- CO1 :: Explain the architecture and routing mechanisms of Next.js applications.
- CO2 :: Apply rendering and data fetching strategies to develop responsive web applications.
- CO3 :: Apply full-stack features using APIs and Server Actions in Next.js.
- CO4 :: Demonstrate authentication and database operations in modern web applications.
- CO5 :: Analyze application performance and deploy production-ready Next.js applications.
- CO6 :: Analyze scalable Next.js application practices based on industry standards.

Unit I

Fundamentals of Next.js : Introduction to Next.js, Features of Next.js, Project Structure, App Router, Development Workflow

Routing and Navigation : File-based Routing, Dynamic Routes, Nested Routes, Layouts, Navigation, Error and Loading UI

Unit II

Rendering Strategies : Server-Side Rendering (SSR), Static Site Generation (SSG), Incremental Static Regeneration (ISR), Client-side Rendering, Rendering Strategy Selection

Data Fetching : Server Components, Client Components, Fetch API, Client-side Fetching, Server-side Data Fetching, Caching and Revalidation

Unit III

API Development : Route Handlers, REST APIs, HTTP Methods, Dynamic Route Handlers, Request and Response Handling

Forms and Server Actions : Server Actions, Form Handling, Form Validation, File Upload, Data Mutations, Optimistic UI Updates

Unit IV

Authentication and Authorization : Authentication Concepts, Session Management, Protected Routes, Authorization, Role-based Access Control

Database Integration : ORM Basics, Database Connectivity, CRUD Operations, Data Validation, Environment Variables

Unit V

Performance and SEO : Image Optimization, Metadata API, Code Splitting, Lazy Loading, SEO Fundamentals

Deployment : Production Build, Environment Configuration, Deployment Platforms, Performance Monitoring, Error Logging

Unit VI

Advanced Next.js Features : Parallel Routes, Proxy, Intercepting Routes, Edge Runtime, Internationalization

Industry Practices : Project Architecture, Component Organization, Security Best Practices, Testing Overview

List of Practicals / Experiments:

Practicals

- Create a Next.js application and explore the App Router-based project structure and development workflow.
- Implement file-based, nested, and dynamic routing with layouts, navigation, loading, and error UI.
- Implement server-side and client-side data fetching and compare different rendering strategies in Next.js.
- Apply caching and revalidation techniques to manage and update application data efficiently.
- Develop REST APIs using Route Handlers and perform operations using different HTTP methods.

- Implement forms with validation, file handling, and data mutations using Server Actions.
- Connect a Next.js application with a database using an ORM and perform CRUD operations.
- Implement authentication, session management, protected routes, and role-based access control.
- Apply image optimization, metadata, code splitting, and lazy loading to improve application performance and SEO.
- Build and deploy a production-ready Next.js application with environment configuration, monitoring, and error logging.

Text Books: 1. MODERN WEB APPLICATIONS WITH NEXT.JS: LEARN ADVANCED TECHNIQUES TO BUILD AND DEPLOY MODERN, SCALABLE AND PRODUCTION READY REACT APPLICATIONS WITH NEXT.JS by SHUBHAM JAIN AND MATHEW DONY, AVA PUBLISHING

References: 1. SCALABLE APPLICATION DEVELOPMENT WITH NESTJS: LEVERAGE REST, GRAPHQL, MICROSERVICES, TESTING, AND DEPLOYMENT FOR SEAMLESS GROWTH by PACIFIQUE LINJANJA, PACKT PUBLISHING